

RATE OF CHANGE

slope = $\Delta y/\Delta x = (y_2 - y_1)/(x_2 - x_1)$
AROC $[a,b] = (f(b) - f(a))/(b - a) = (f(x) - f(a))/(x - a) \rightarrow$ secant slope
IROC @a = $\lim_{h \rightarrow 0} (f(a+h) - f(a))/h = \lim_{x \rightarrow a} (f(x) - f(a))/(x - a) \rightarrow$ tangent slope
 $f'(x) = \lim_{h \rightarrow 0} (f(x+h) - f(x))/h$ **$f'(a) = m$**
 $f', y', dy/dx, d/dx f|_{x=a}$ (nDeriv)
table: $f'(x) \approx (f(b) - f(a))/(b - a)$, points around x

LINES

$y - y_1 = m(x - x_1)$
Tangent @a: $y - f(a) = f'(a)(x - a)$
Normal: slope = **$-1/f'(a)$** (neg. reciprocal)
Horizontal tangent: $f'(x) = 0$

DIFFERENTIABLE?

NOT at: jump/hole, corner/cusp, vertical tangent
Diff $\Rightarrow$ continuous (T); cont $\Rightarrow$ diff (F)
Piecewise @c: pieces equal at c, AND derivatives equal at c

ROOTS / POWERS

$\sqrt{x} = x^{1/2}$; $\sqrt[n]{x^m} = x^{(m/n)}$; $1/x^n = x^{-n}$
 $x^a \cdot x^b = x^{a+b}$; $x^a/x^b = x^{a-b}$; $(x^a)^b = x^{ab}$
Rewrite as powers BEFORE deriving

DERIVATIVE RULES

$d/dx c = 0$; $d/dx cx = c$; $(cu)' = c \cdot u'$
 $(u \pm v)' = u' \pm v'$

Power: $(x^n)' = n \cdot x^{n-1}$; $(\sqrt{x})' = 1/(2\sqrt{x})$;
 $(1/x)' = -1/x^2$

Product: $(fg)' = f \cdot g' + g \cdot f'$
Quotient: $(f/g)' = (g \cdot f' - f \cdot g')/g^2$ "ho d hi - hi d ho / ho ho"

EXP / LOG

$(e^x)' = e^x$; $(a^x)' = a^x \ln a$
 $(\ln x)' = 1/x$; $(\log_a x)' = 1/(x \ln a)$
 $\ln 1 = 0$; $\ln 0 = \text{DNE}$; $\ln e = 1$; $e^0 = 1$;
 $e^{(\ln a)} = a$; $\ln(e^a) = a$

TRIG DERIVATIVES

$\sin \rightarrow \cos$; $\cos \rightarrow -\sin$; $\tan \rightarrow \sec^2$; $\cot \rightarrow -\csc^2$; $\sec \rightarrow \sec \cdot \tan$; $\csc \rightarrow -\csc \cdot \cot$
 $\csc = 1/\sin$; $\sec = 1/\cos$; $\tan = \sin/\cos$;
 $\cot = \cos/\sin$

TRIG VALUES (DNE = undefined)

θ	$\sin$	$\cos$	$\tan$	$\csc$	$\sec$	$\cot$
0	0	1	0	DNE	1	DNE
$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$	2	$2\sqrt{3}/3$	$\sqrt{3}$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1	$\sqrt{2}$	$\sqrt{2}$	1
$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$	$2\sqrt{3}/3$	2	$\sqrt{3}/3$
$\pi/2$	1	0	DNE	1	DNE	0
$2\pi/3$	$\sqrt{3}/2$	$-1/2$	$-\sqrt{3}$	$2\sqrt{3}/3$	-2	$-\sqrt{3}/3$
$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	-1	$\sqrt{2}$	$-\sqrt{2}$	-1
$5\pi/6$	$1/2$	$-\sqrt{3}/2$	$-\sqrt{3}/3$	2	$-2\sqrt{3}/3$	$-\sqrt{3}$
π	0	-1	0	DNE	-1	DNE
$7\pi/6$	$-1/2$	$-\sqrt{3}/2$	$\sqrt{3}/3$	-2	$-2\sqrt{3}/3$	$\sqrt{3}$
$5\pi/4$	$-\sqrt{2}/2$	$-\sqrt{2}/2$	1	$-\sqrt{2}$	$-\sqrt{2}$	1
$4\pi/3$	$-\sqrt{3}/2$	$-1/2$	$\sqrt{3}$	$-2\sqrt{3}/3$	-2	$\sqrt{3}/3$
$3\pi/2$	-1	0	DNE	-1	DNE	0
$5\pi/3$	$-\sqrt{3}/2$	$1/2$	$-\sqrt{3}$	$-2\sqrt{3}/3$	2	$-\sqrt{3}/3$
$7\pi/4$	$-\sqrt{2}/2$	$\sqrt{2}/2$	-1	$-\sqrt{2}$	$\sqrt{2}$	-1
$11\pi/6$	$-1/2$	$\sqrt{3}/2$	$-\sqrt{3}/3$	-2	$2\sqrt{3}/3$	$-\sqrt{3}$
2π	0	1	0	DNE	1	DNE